

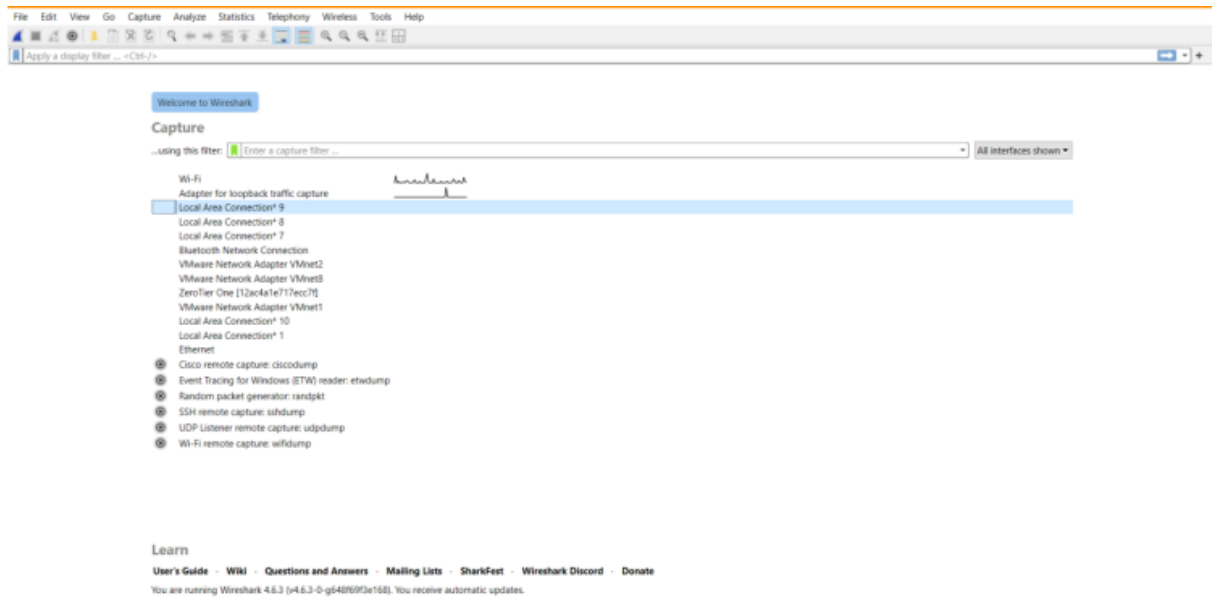
Advanced Computer Networks – PCS223

Assignment 2

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Subgroup – CS4

Roll No. – 8025320107



Screenshot showing Wireshark Home screen showing all interfaces and the active interface (Wi-Fi)

1. To capture and analyze ICMP packets (Ping request and reply)

Capturing from Wi-Fi

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
33890	2104.889882	104.196.0.153	192.168.1.9	ICMP	70	Destination unreachable (Port unreachable)
33918	2105.487238	104.196.0.153	192.168.1.9	ICMP	70	Destination unreachable (Port unreachable)
33927	2106.695621	104.196.0.153	192.168.1.9	ICMP	70	Destination unreachable (Port unreachable)
33947	2108.575557	104.196.0.153	192.168.1.9	ICMP	70	Destination unreachable (Port unreachable)
35353	2217.858954	192.168.1.1	192.168.1.9	ICMP	76	Destination unreachable (Port unreachable)
44928	2523.149478	192.168.1.1	192.168.1.9	ICMP	76	Destination unreachable (Port unreachable)
50790	2829.425452	192.168.1.1	192.168.1.9	ICMP	76	Destination unreachable (Port unreachable)
83130	3133.781818	192.168.1.1	192.168.1.9	ICMP	76	Destination unreachable (Port unreachable)
85695	3260.185576	192.168.1.9	142.250.192.206	ICMP	74	Echo (ping) request id=0x0001, seq=11/2816, ttl=128 (reply in 85696)
85696	3260.197718	142.250.192.206	192.168.1.9	ICMP	74	Echo (ping) reply id=0x0001, seq=11/2816, ttl=119 (request in 85695)
85703	3261.194224	192.168.1.9	142.250.192.206	ICMP	74	Echo (ping) request id=0x0001, seq=12/3072, ttl=128 (reply in 85704)
85704	3261.206757	142.250.192.206	192.168.1.9	ICMP	74	Echo (ping) reply id=0x0001, seq=12/3072, ttl=119 (request in 85703)
85723	3262.204261	192.168.1.9	142.250.192.206	ICMP	74	Echo (ping) request id=0x0001, seq=13/3328, ttl=128 (reply in 85724)
85724	3262.217717	142.250.192.206	192.168.1.9	ICMP	74	Echo (ping) reply id=0x0001, seq=13/3328, ttl=119 (request in 85723)
85729	3263.227535	192.168.1.9	142.250.192.206	ICMP	74	Echo (ping) request id=0x0001, seq=14/3584, ttl=128 (reply in 85730)
85730	3263.239864	142.250.192.206	192.168.1.9	ICMP	74	Echo (ping) reply id=0x0001, seq=14/3584, ttl=119 (request in 85729)

< >

Frame 18006: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface \Device\NPF...
 Ethernet II, Src: zte_e0:60:9c (d4:b7:09:e0:60:9c), Dst: Intel_19:3a:b1 (8c:c6:81:19:3a:b1)
 Destination: Intel_19:3a:b1 (8c:c6:81:19:3a:b1)
 Source: zte_e0:60:9c (d4:b7:09:e0:60:9c)
 Type: 802.1Q Virtual LAN (0x8100)
 [Stream index: 0]
 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
 Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.9
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
 Total Length: 58
 Identification: 0x447b (17531)
 000. = Flags: 0x0
 ...0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 64
 Protocol: ICMP (1)
 Header Checksum: 0xb22d (validation disabled)
 Payload: ICMP Echo (ping) request

0000 8c c6 81 19 3a b1 d4 b7 09 e0 60 9c 81 00 00 00
 0010 08 00 45 c0 00 3a d4 7b 00 00 40 01 b2 2d c0 a8 --E--D--@---
 0020 01 01 c0 a8 01 09 03 03 80 73 00 00 00 45 00s---E-
 0030 00 1e 76 41 00 00 80 11 41 33 c0 a8 01 09 c0 a8 --VA---A3-----
 0040 01 01 e5 fd 14 e7 00 0a 81 9a 00 00--

Screenshot showing ICMP Packets Captured by pinging google.com

Capturing from Wi-Fi

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Apply a display filter ... <Ctrl>/>

No.	Time	Source	Destination	Protocol	Length	Info
3417	267.027255	192.168.1.9	142.250.193.67	QUIC	75	Protected Payload (QPO), DCID=f4eb850a3c27d5de
3418	267.095953	142.250.193.67	192.168.1.9	QUIC	181	Protected Payload (QPO)
3419	267.095989	192.168.1.9	142.250.193.67	QUIC	75	Protected Payload (QPO), DCID=f4eb850a3c27d5de
3420	267.060864	142.250.193.67	192.168.1.9	QUIC	64	Protected Payload (QPO)
3421	267.061100	192.168.1.9	142.250.193.67	QUIC	75	Protected Payload (QPO), DCID=f4eb850a3c27d5de
3422	267.095618	142.250.193.67	192.168.1.9	QUIC	66	Protected Payload (QPO)
3423	267.289145	192.168.1.9	172.217.27.174	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 3424)
3424	267.303549	172.217.27.174	192.168.1.9	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=119 (request in 3423)
3425	269.245315	zte_e0:60:9c	Intel_19:3a:b1	ARP	42	Who has 192.168.1.9? Tell 192.168.1.1
3426	269.245346	Intel_19:3a:b1	zte_e0:60:9c	ARP	42	192.168.1.9 is at 8c:c6:81:19:3a:b1
3427	269.342278	192.168.1.9	103.195.103.66	UDP	179	9993 → 9993 Len=137
3428	269.342378	192.168.1.9	103.195.103.66	UDP	179	44168 → 9993 Len=137
3429	269.342414	192.168.1.9	103.195.103.66	UDP	179	44169 → 9993 Len=137
3430	269.342503	192.168.1.9	79.127.159.187	UDP	179	9993 → 9993 Len=137
3431	269.342537	192.168.1.9	79.127.159.187	UDP	179	44168 → 9993 Len=137
3432	269.342576	192.168.1.9	79.127.159.187	UDP	179	44169 → 9993 Len=137
3433	269.342667	192.168.1.9	84.17.53.155	UDP	179	9993 → 9993 Len=137

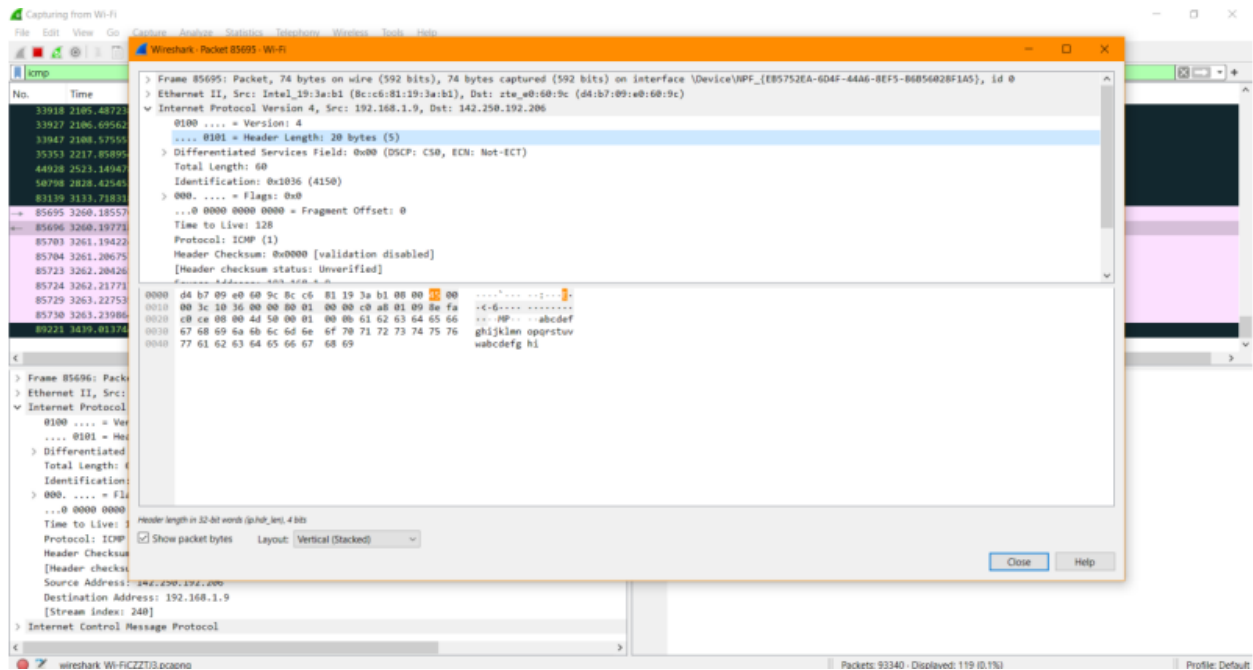
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Frame 1: Packet, 591 bytes on wire (4728 bits), 591 bytes captured (4728 bits) on interface \Device\NPF...
 Ethernet II, Src: Intel_19:3a:b1 (8c:c6:81:19:3a:b1), Dst: zte_e0:60:9c (d4:b7:09:e0:60:9c)
 Internet Protocol Version 4, Src: 192.168.1.9, Dst: 142.250.193.46
 User Datagram Protocol, Src Port: 57654, Dst Port: 443
 Data (549 bytes)

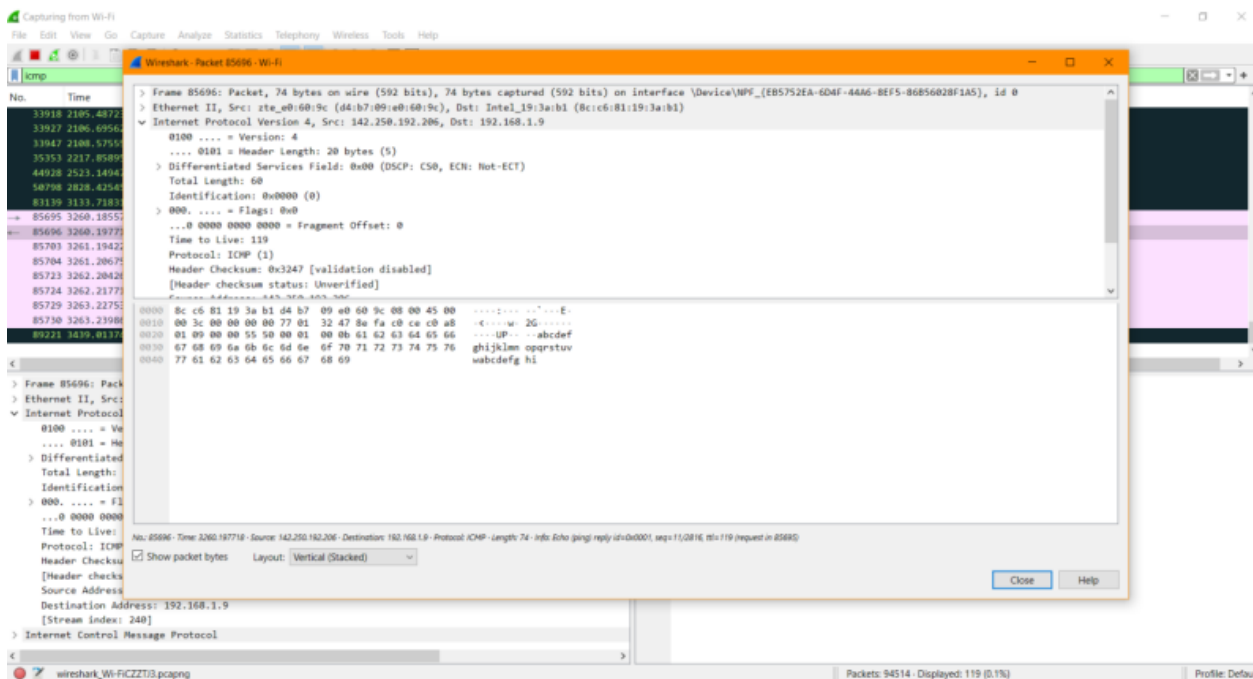
0000 d4 b7 09 e0 60 9c 8c c6 81 19 3a b1 00 00 45 00E-
 0010 02 41 b6 09 40 00 00 51 00 00 c0 a8 01 09 0e fa -A-@-----
 0020 c1 2a e1 36 01 bb 02 2d 14 19 4a fc 63 0e 37 bb -.-G---B-c-?
 0030 52 d4 b3 8d 8b d7 c8 06 ad 8a ff 20 22 39 e9 eb R-----"9-
 0040 61 30 c6 78 3d 51 00 33 c7 19 08 7e 04 70 44 15 a0 -wQ-3---d0-
 0050 36 ae da 83 d7 38 75 98 5c f7 05 90 10 25 d3 22 G---B-[-S-
 0060 66 b0 f3 3b c6 da 21 e6 99 97 4a 7a 70 ff 06 e2 f-[-]-0p-
 0070 ec 05 09 6c 67 00 ea fd 81 17 2a 80 9d d3 6f d3 -lig-+---p-
 0080 90 be 1a 16 c2 f9 14 1c 21 6c 30 6c ac f4 26 8d1101-B
 0090 95 f9 d3 c8 3c 03 f3 25 b2 b5 7a 6b f9 43 ef 27 ---G-X-ak-C-
 00a0 70 10 2f 93 d2 dc 7c e0 20 6f a2 67 79 8e 63 9d w/-[-]-o-g-c-
 00b0 0f 97 30 e9 40 75 74 aa 4c b0 6f c4 68 ab 2a 06 -B-but-L-o-h-
 00c0 35 b0 4c 52 c9 b0 80 4c fb 79 a6 d2 10 40 38 34 5-LR---L-y-HB4
 00d0 57 c7 e6 52 d2 00 30 c4 b7 3d 69 ea 2a 7d 4a 7e w-R-B-[-]-*]-
 00e0 4f d3 4c ac ff c5 a7 25 e6 5d a1 bb 18 2a 94 25 O-L-[-S-]-X
 00f0 c2 f8 76 f1 68 47 f4 75 2e de a3 a4 8f 6c 99 07 -wHG-u-1-
 0100 6e 88 b4 74 72 b1 ce 58 d1 ab 30 c3 dc 59 00 6f -m-r-X-0-Y-o
 0110 6d 09 a8 64 da 7c fd 0f 34 9f b4 dc f9 5b 96 26 -d-]-4-[-L-
 0120 a8 cb 69 b7 f3 a4 22 4c 00 f3 c9 47 2f 7d 75 00 -i-[-]-c-6-3-
 0130 b5 0b 51 5a cf 98 36 7c b9 28 57 cf a0 1f 58 3c -Q-b-6-[-]W-H-c
 0140 a7 91 66 84 62 0a 35 04 02 b8 2f fa 9a 83 d3 24 -F-b-5-[-]-S-
 0150 bd 38 97 71 86 60 c4 05 f1 47 40 b7 3a 06 77 39 -B-q-[-]G->-w9
 0160 ac 4f f7 7b be a0 21 de b4 b9 d7 10 ee 4a 89 6e -D-[-]-[-]->-n

Wi-Fi - live capture in progress... Packets: 3712 Profile: Default

Screenshot showing ICMP Packets Captured by pinging google.com



Screenshot showing ICMP Packet 85695 details (Ping Request)



Screenshot showing ICMP Packet 85696 details (Ping Reply)

Observation and Analysis of ICMP Packet 85695 -

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 142.250.192.206
- Packet Length: 74 Bytes

Observation and Analysis of ICMP Packet 85696 -

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol
- Source IP Address: 142.250.192.206
- Destination IP Address: 192.168.1.9
- Packet Length: 74 Bytes

2. To analyze TCP packets and observe the TCP three-way handshake

Wi-Fi

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tcpstream eq 155

No.	Time	Source	Destination	Protocol	Length	Info
148816	851.687749	172.16.69.133	142.250.182.234	TCP	66	63331 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
148836	851.740898	142.250.182.234	172.16.69.133	TCP	66	443 → 63331 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
148844	851.741272	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
148852	851.744521	172.16.69.133	142.250.182.234	TLSv1.3	1824	Client Hello (SNI=ogads-pa.clients6.google.com)
148857	851.758195	142.250.182.234	172.16.69.133	TCP	60	443 → 63331 [ACK] Seq=1 Ack=1771 Win=268032 Len=0
148858	851.758195	142.250.182.234	172.16.69.133	TLSv1.3	1466	Server Hello, Change Cipher Spec
148859	851.758195	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=1413 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148860	851.758303	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=2825 Win=131072 Len=0
148893	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=2825 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148894	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=4237 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148895	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=5649 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148896	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=7061 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148897	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=8473 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148904	851.799244	142.250.182.234	172.16.69.133	TLSv1.3	1108	Application Data
148907	851.799322	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=9885 Win=131072 Len=0
148908	851.799474	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=10939 Win=130048 Len=0
148909	851.799940	172.16.69.133	142.250.182.234	TLSv1.3	144	Change Cipher Spec, Application Data

> Ethernet II, Src: Intel_19:3a:b1 (8c:c6:81:19:3a:b1), Dst: BrocadeComm_90:80:78 (74:8e:f8:90:80:78)

> Internet Protocol Version 4, Src: 172.16.69.133, Dst: 142.250.182.234

> Transmission Control Protocol, Src Port: 63331, Dst Port: 443, Seq: 0, Len: 0

Source Port: 63331

Destination Port: 443

[Stream index: 155]

[Stream Packet Number: 1]

> [Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 295960920

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 0

Acknowledgment number (raw): 0

1000 = Header Length: 32 bytes (8)

> Flags: 0x002 (SYN)

Window: 64240

[Calculated window size: 64240]

Checksum: 0x37a5 [unverified]

Source Port (tcp.srcport), 2 bytes

Packets: 149179 - Displayed: 34 (0.0%) - Dropped: 0 (0.0%)

Screenshot showing TCP Packet (SYN)

Wi-Fi

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tcpstream eq 155

No.	Time	Source	Destination	Protocol	Length	Info
148816	851.687749	172.16.69.133	142.250.182.234	TCP	66	63331 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
148836	851.740898	142.250.182.234	172.16.69.133	TCP	66	443 → 63331 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
148844	851.741272	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
148852	851.744521	172.16.69.133	142.250.182.234	TLSv1.3	1824	Client Hello (SNI=ogads-pa.clients6.google.com)
148857	851.758195	142.250.182.234	172.16.69.133	TCP	60	443 → 63331 [ACK] Seq=1 Ack=1771 Win=268032 Len=0
148858	851.758195	142.250.182.234	172.16.69.133	TLSv1.3	1466	Server Hello, Change Cipher Spec
148859	851.758195	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=1413 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148860	851.758303	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=2825 Win=131072 Len=0
148893	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=2825 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148894	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=4237 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148895	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=5649 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148896	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=7061 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148897	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=8473 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148904	851.799244	142.250.182.234	172.16.69.133	TLSv1.3	1108	Application Data
148907	851.799322	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=9885 Win=131072 Len=0
148908	851.799474	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=10939 Win=130048 Len=0
148909	851.799940	172.16.69.133	142.250.182.234	TLSv1.3	144	Change Cipher Spec, Application Data

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 2124938681

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 295960921

1000 = Header Length: 32 bytes (8)

> Flags: 0x012 (SYN, ACK)

Window: 65535

[Calculated window size: 65535]

Checksum: 0xaf3b [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

> Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted,

> [Timestamps]

> [SEQ/ACK analysis]

[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

Source Port (tcp.srcport), 2 bytes

Packets: 149179 - Displayed: 34 (0.0%) - Dropped: 0 (0.0%)

Screenshot showing TCP Packet (SYN, ACK)

Wireshark interface showing a list of captured packets. The selected packet (No. 148904) is a TCP ACK segment from 172.16.69.133 to 142.250.182.234.

No.	Time	Source	Destination	Protocol	Length	Info
148816	0.01687749	172.16.69.133	142.250.182.234	TCP	66	63331 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
148836	0.01740898	142.250.182.234	172.16.69.133	TCP	66	443 → 63331 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
148844	0.01741272	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1 Ack=1 Min=131072 Len=0
148852	0.01744521	172.16.69.133	142.250.182.234	TLv1.3	1824	Client Hello (SHA256, TLSv1.3)
148857	0.01758195	142.250.182.234	172.16.69.133	TCP	60	443 → 63331 [ACK] Seq=1 Ack=1771 Win=268032 Len=0
148858	0.01758195	142.250.182.234	172.16.69.133	TLv1.3	1466	Server Hello, Change Cipher Spec
148859	0.01758195	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=1413 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148860	0.01758303	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=2825 Win=131072 Len=0
148893	0.01799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=2825 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148894	0.01799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=4237 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148895	0.01799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=5649 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148896	0.01799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=7061 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148897	0.01799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=8473 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148904	0.01799244	142.250.182.234	172.16.69.133	TLv1.3	1188	Application Data
148907	0.01799322	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=8885 Win=131072 Len=0
148908	0.01799474	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=10939 Win=138048 Len=0
148909	0.01799540	172.16.69.133	142.250.182.234	TLv1.3	144	Change Cipher Spec, Application Data

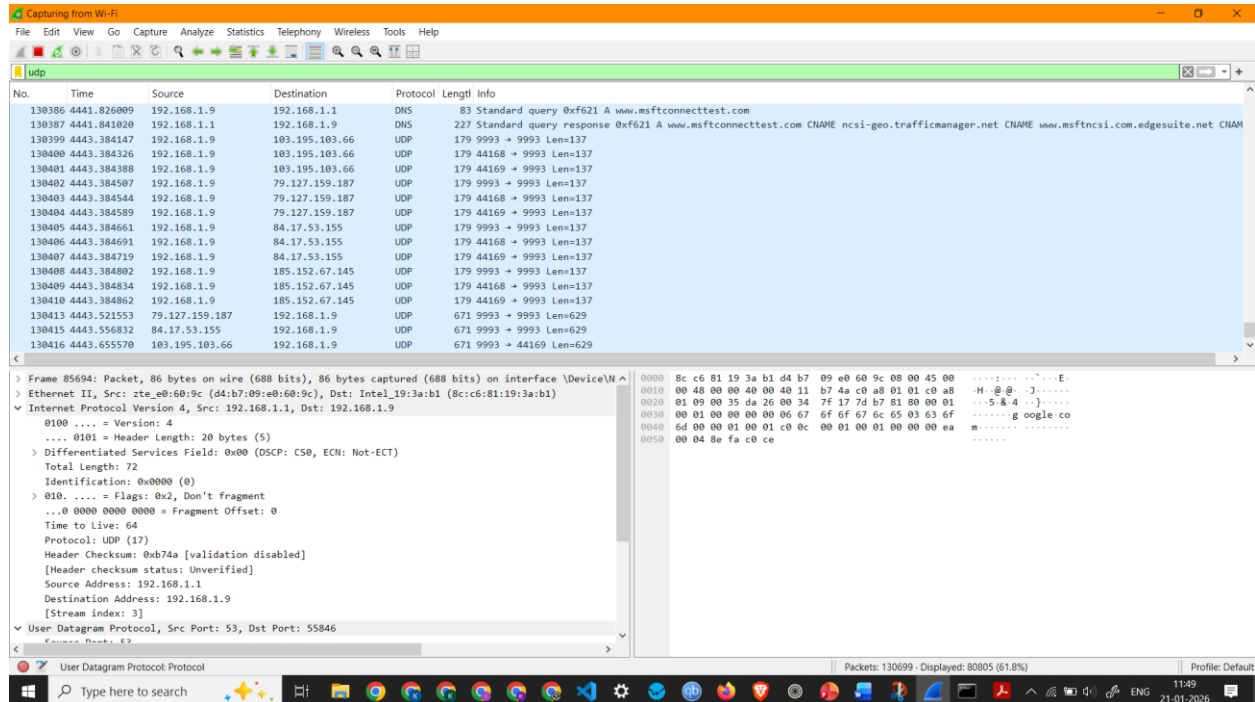
Packet 148904 Details:

- [Stream Packet Number: 3]
- [Conversation completeness: Incomplete, DATA (15)]
- [TCP Segment Len: 0]
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 295960921
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)
- Acknowledgment number (raw): 2124938682
- 0101 = Header Length: 20 bytes (5)
- Flags: 0x010 (ACK)**
- Window: 512
- [Calculated window size: 131072]
- [Window size scaling factor: 256]
- Checksum: 0x3795 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- [Timestamps]
- [SEQ/ACK analysis]

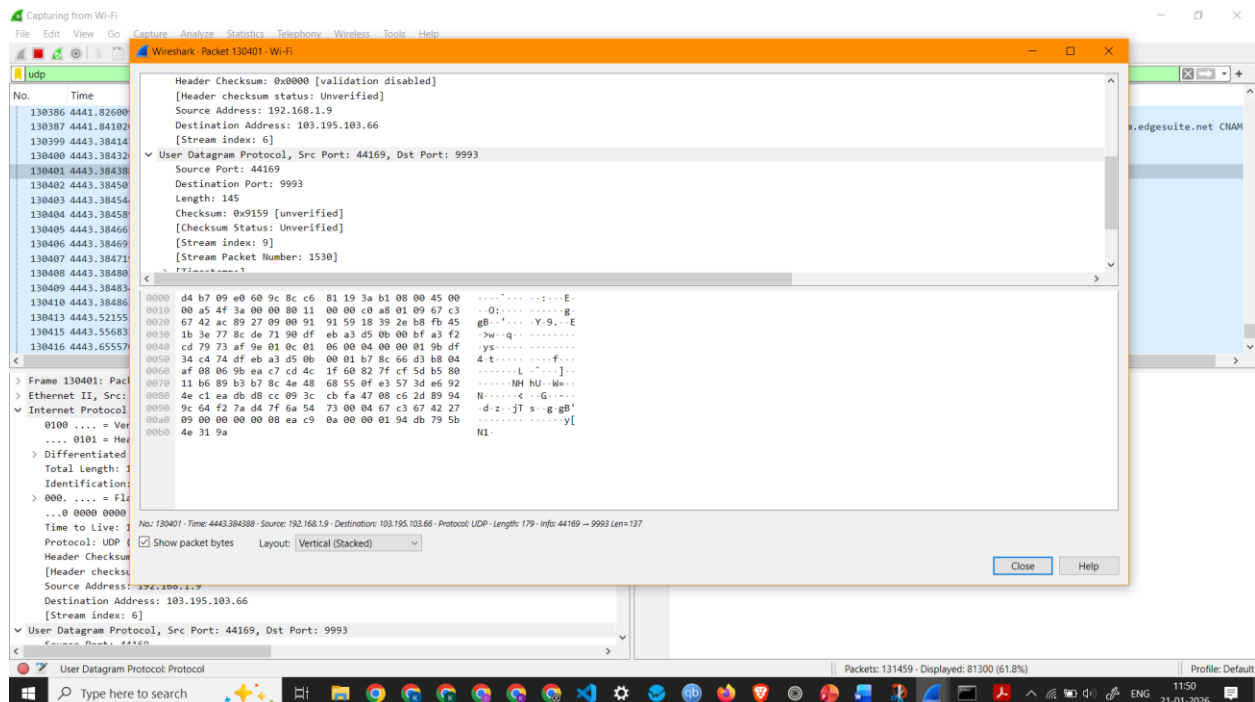
Raw packet data (hex): 74 Be f8 90 80 78 8c c6 81 19 3a b1 00 00 45 00 00 28 72 5e 40 00 00 06 00 00 ac 10 45 85 Be fa 00 20 b6 ea f7 63 01 bb 11 a4 01 59 7e a7 fd ba 02 00 37 95 00 00

Screenshot showing TCP Packet (ACK)

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture

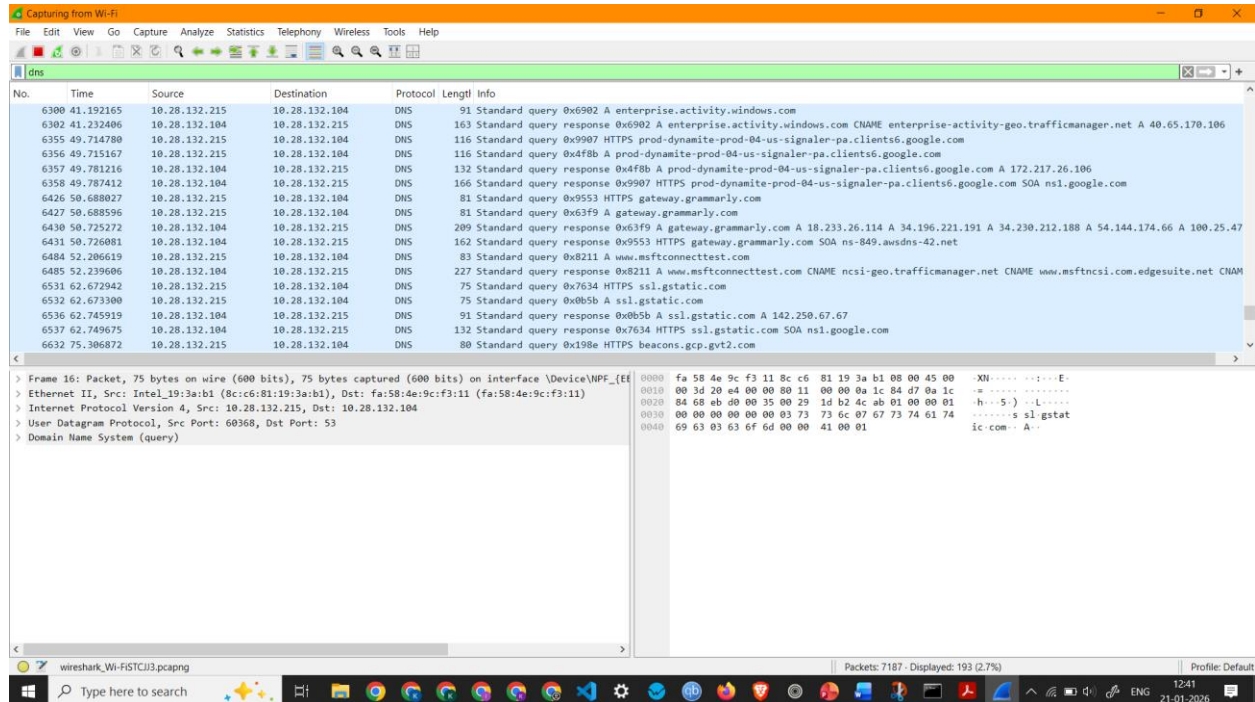


Screenshot showing details of a UDP Packet

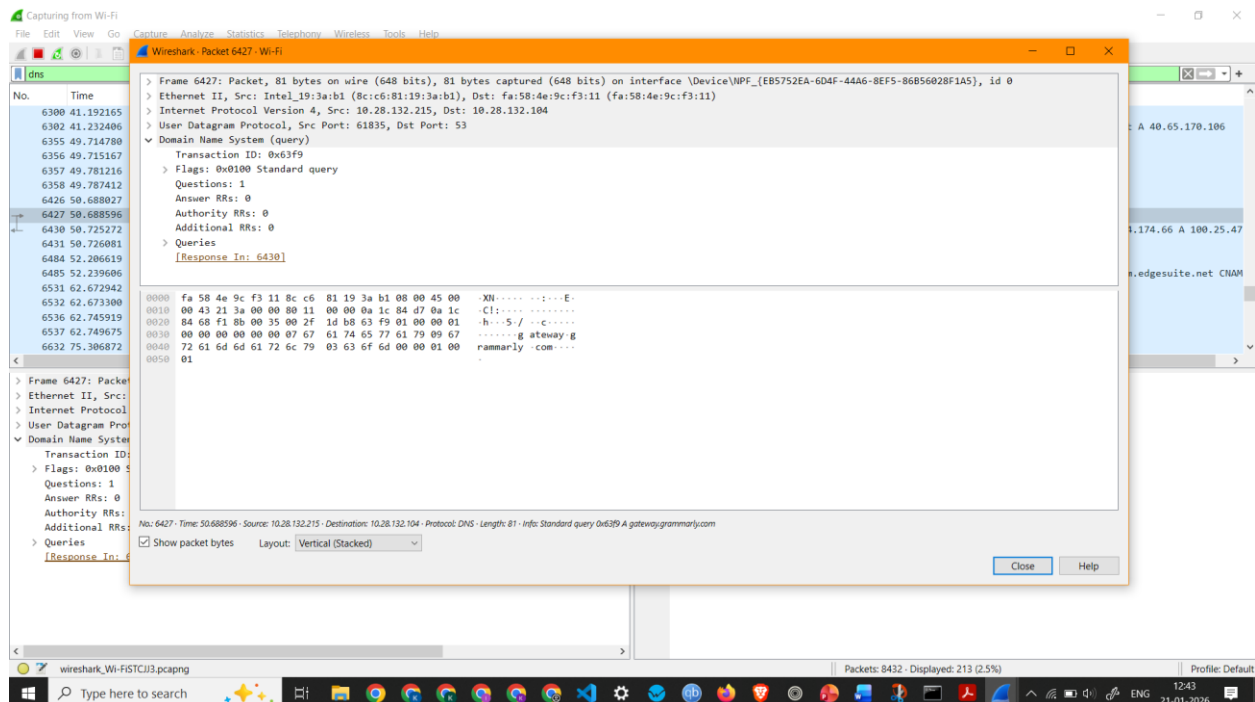
Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

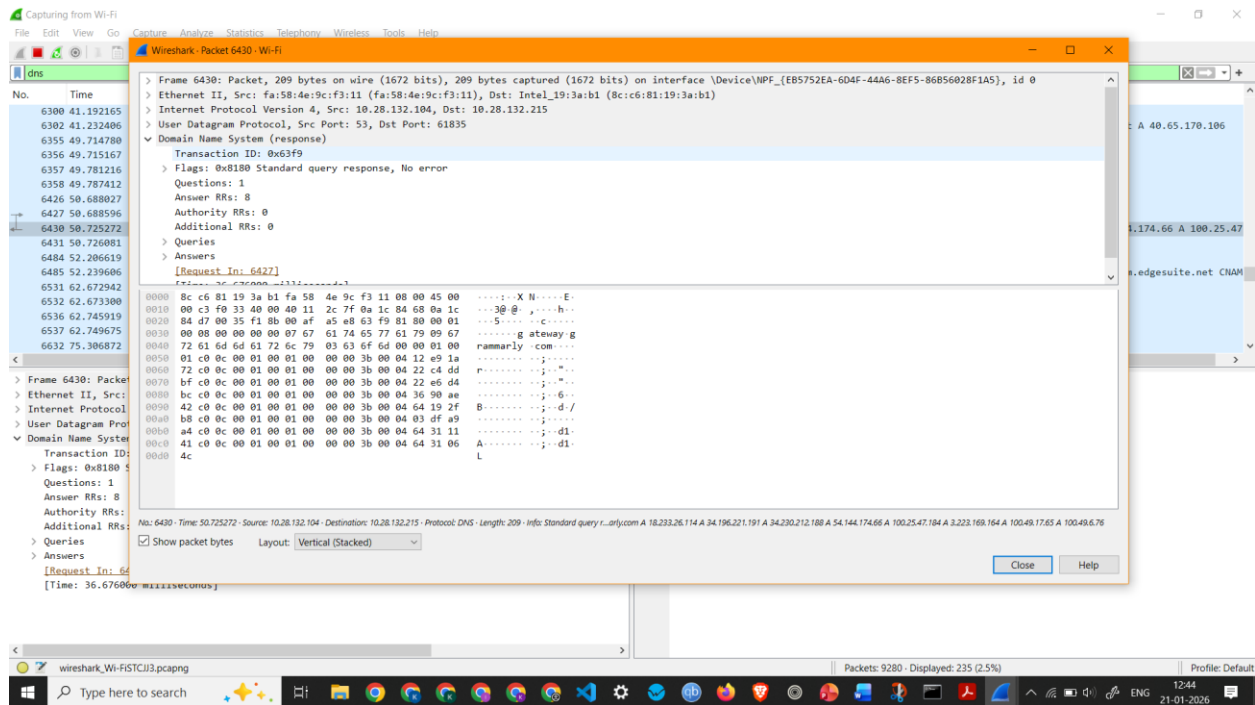
4. To capture and analyze DNS query and response packets



Screenshot showing DNS Packets Capture



Screenshot showing details of a DNS Packet 6427 (Standard query request)



Screenshot showing details of a DNS Packet 6430 (Standard query response)

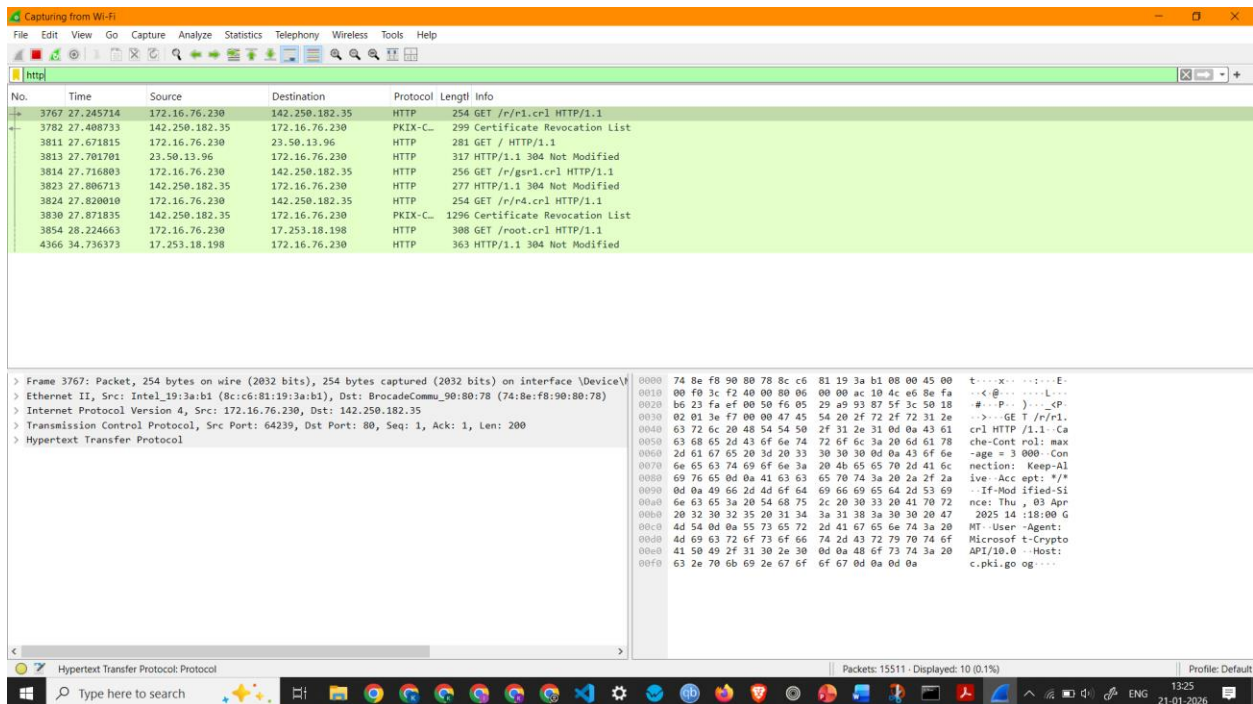
Observation: DNS Packet 6427 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.215
- Destination IP Address: 10.28.132.104
- Packet Length: 81 Bytes
- Queries - gateway.grammarly.com: type A, class IN

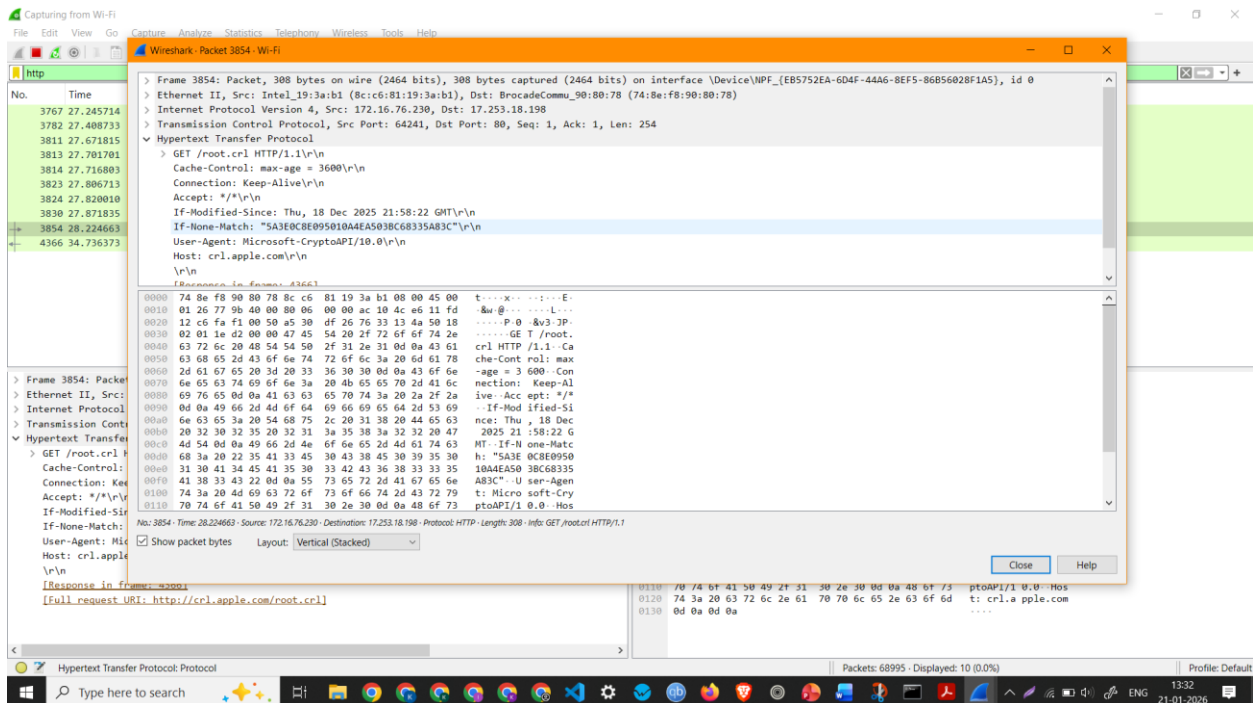
Observation: DNS Packet 6430 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.104
- Destination IP Address: 10.28.132.215
- Packet Length: 209 Bytes

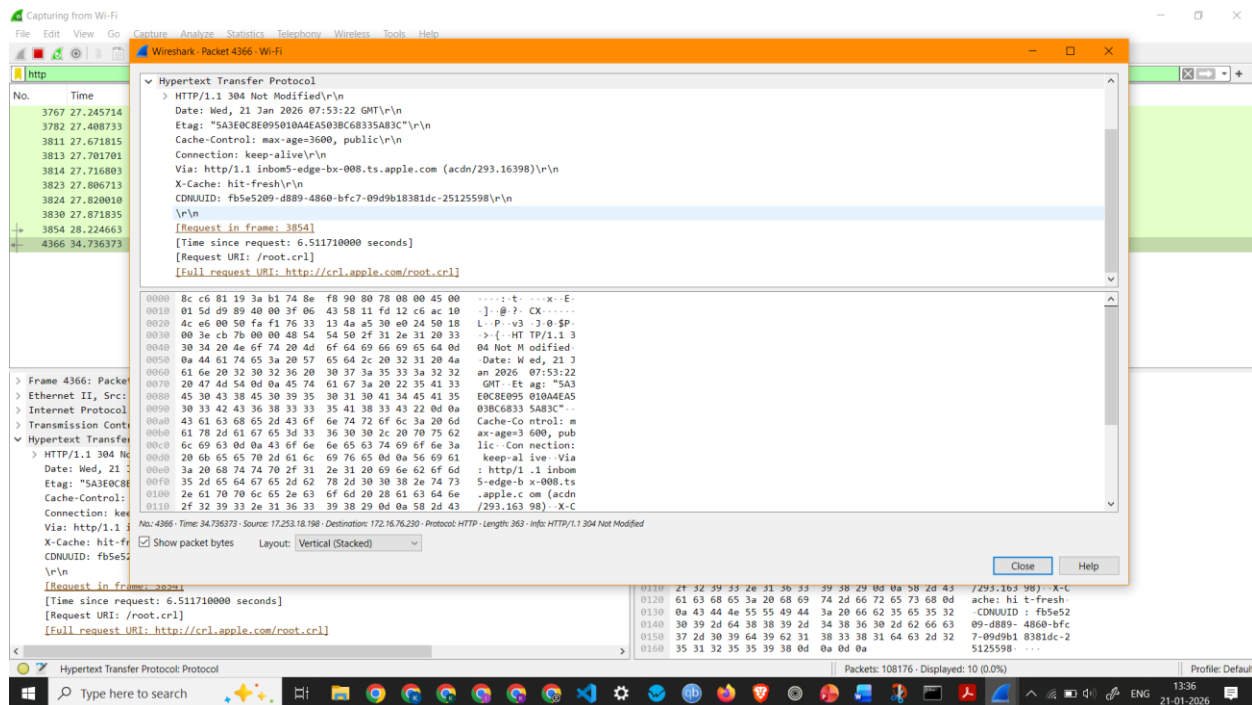
5. To capture and analyze HTTP packets



Screenshot showing HTTP Packets Capture



Screenshot showing details of a HTTP Packet (HTTP Request)



Screenshot showing details of a HTTP Packet (HTTP Response)

Observation: HTTP Request 3854 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.16.76.230
- Destination IP Address: 17.253.18.198
- Packet Length: 308 Bytes

Observation: HTTP Response 4366 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 17.253.18.198
- Destination IP Address: 172.16.76.230
- Packet Length: 363 Bytes

6. To analyze ARP request and reply packets

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for various functions like opening files, saving, and analyzing. The packet list pane on the left shows a list of captured packets, with packet 21 selected. The packet details pane on the right shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and ARP. The packet bytes pane on the right shows the raw data of the selected packet in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
4	0.103477	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.74.134? Tell 172.16.64.1
21	0.196499	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.73.115? Tell 172.16.64.1
22	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.77.107? Tell 172.16.64.1
23	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.93.88? Tell 172.16.64.1
24	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.75.224? Tell 172.16.64.1
25	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.95.46? Tell 172.16.64.1
26	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.68.235? Tell 172.16.64.1
27	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.74.134? Tell 172.16.64.1
28	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.73.129? Tell 172.16.64.1
29	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.69.106? Tell 172.16.64.1
30	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.74.102? Tell 172.16.64.1
31	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.91.60? Tell 172.16.64.1
32	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.81.154? Tell 172.16.64.1
33	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.92.219? Tell 172.16.64.1
34	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.94.96? Tell 172.16.64.1
35	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.91.186? Tell 172.16.64.1
36	0.206316	BrocadeComm_98:80:...	Broadcast	ARP	60	Who has 172.16.83.209? Tell 172.16.64.1

Frame 21: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{...}

Section number: 1

Interface id: 0 (\Device\NPF_{E85752EA-604F-44A6-8EF5-86B56028F1A5})

Encapsulation type: Ethernet (1)

Arrival Time: Apr 1, 2026 13:25:54.747861000 India Standard Time

UTC Arrival Time: Apr 1, 2026 07:55:54.747861000 UTC

Epoch Arrival Time: 1775030154.747861000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 91.281000 milliseconds]

[Time delta from previous displayed frame: 93.022000 milliseconds]

[Time since reference or first frame: 196.499000 milliseconds]

Frame Number: 21

Frame Length: 60 bytes (480 bits)

Capture Length: 60 bytes (480 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:arp]

Character encoding: ASCII (0)

Address Resolution Protocol: Protocol

Packets: 149179 - Displayed: 38629 (25.9%) - Dropped: 0 (0.0%)

Screenshot showing ARP Packets Capture

7. To identify source and destination MAC and IP addresses in captured packets

The screenshot displays the Wireshark network traffic capture interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for packet capture and analysis. The main display area shows a list of captured packets, with the selected packet (No. 38328) highlighted in green. The packet details pane on the left shows the selected packet's structure, including the Internet Protocol Version 4 header and the Hypertext Transfer Protocol header. The packet bytes pane on the right shows the raw data of the selected packet, including the IP header and the HTTP GET request body.

No.	Time	Source	Destination	Protocol	Length	Info
38328	14.136535	172.16.69.133	34.223.124.45	HTTP	481	GET / HTTP/1.1
80726	27.793747	172.16.69.133	34.223.124.45	HTTP	507	GET / HTTP/1.1
321033	122.067090	172.16.69.133	23.11.39.161	HTTP	288	GET /MFewTzBNMEswSTA3BgUrDgMCGGUABBTjrydRyt%2BApF3GSPypfHBxR5XtQQUs9tIpMhxdIUkHMEWNPYim8SBYCEAI5PUjXAKJaf
321464	122.188176	23.11.39.161	172.16.69.133	OCSP	538	Response
350216	134.149918	172.16.69.133	104.18.27.120	HTTP	129	GET / HTTP/1.1
351492	134.773127	104.18.27.120	172.16.69.133	HTTP	61	HTTP/1.1 200 OK (text/html)

Packet details for the selected packet (No. 38328):

- [Stream index: 99]
- Internet Protocol Version 4, Src: 172.16.69.133, Dst: 34.223.124.45
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 467
 - Identification: 0x14da (5338)
 - > 010. = Flags: 0x2, Don't fragment
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 128
 - Protocol: TCP (6)
 - Header Checksum: 0x0000 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 172.16.69.133
 - Destination Address: 34.223.124.45
 - [Stream index: 790]
- Transmission Control Protocol, Src Port: 63964, Dst Port: 80, Seq: 1, Ack: 1, Len: 427
- Hypertext Transfer Protocol

Packet bytes for the selected packet (No. 38328):

0010 01 d3 14 da 40 00 80 06 00 00 ac 10 45 85 22 d1@... ..E..

0020 7c 20 f9 dc 00 50 ad c6 eb 9d 2c d0 1e 6e 50 18 ..g.P... ..nP

0030 02 01 92 67 00 00 47 45 54 20 2f 20 48 54 50 18 ..g..GE T / HTTP

0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6e 65 76 65 /1.1..Ho st: neve

0050 72 73 73 6c 2e 63 6f 6d 0d 0a 43 6f 6e 6e 65 63 rssl.com ..Connec

0060 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 tion: ke ep-alive

0070 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75 ..Upgrad e-Insecu

0080 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d 0a re-Reqe sts: 1..

0090 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 User-Age nt: Mozi

00a0 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 lla/5.0 (Windows

00b0 20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b NT 10.0 ; Win64;

00c0 20 78 36 34 29 20 41 70 70 6c 65 57 65 62 4b 69 x64) Ap pleWebKi

00d0 74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c t/537.36 (KHTML,

00e0 20 6c 69 6b 65 20 47 65 63 6b 6f 29 20 43 68 72 like Ge cko) Chr

00f0 6f 6d 65 2f 31 34 36 2e 30 2e 30 2e 30 20 53 61 ome/146.0.0.0 Sa

0100 66 61 72 69 2f 35 33 37 2e 33 36 0d 0a 41 63 63 fari/537.36-Acc

0110 65 70 74 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61 ept: tex t/html,a

0120 70 70 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c pplicati on/xhtml

0130 2b 78 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e +xml,app lication

0140 2f 78 6d 6c 3b 71 3d 30 2e 39 2c 69 6d 61 67 65 /xml;q=0 .9,image

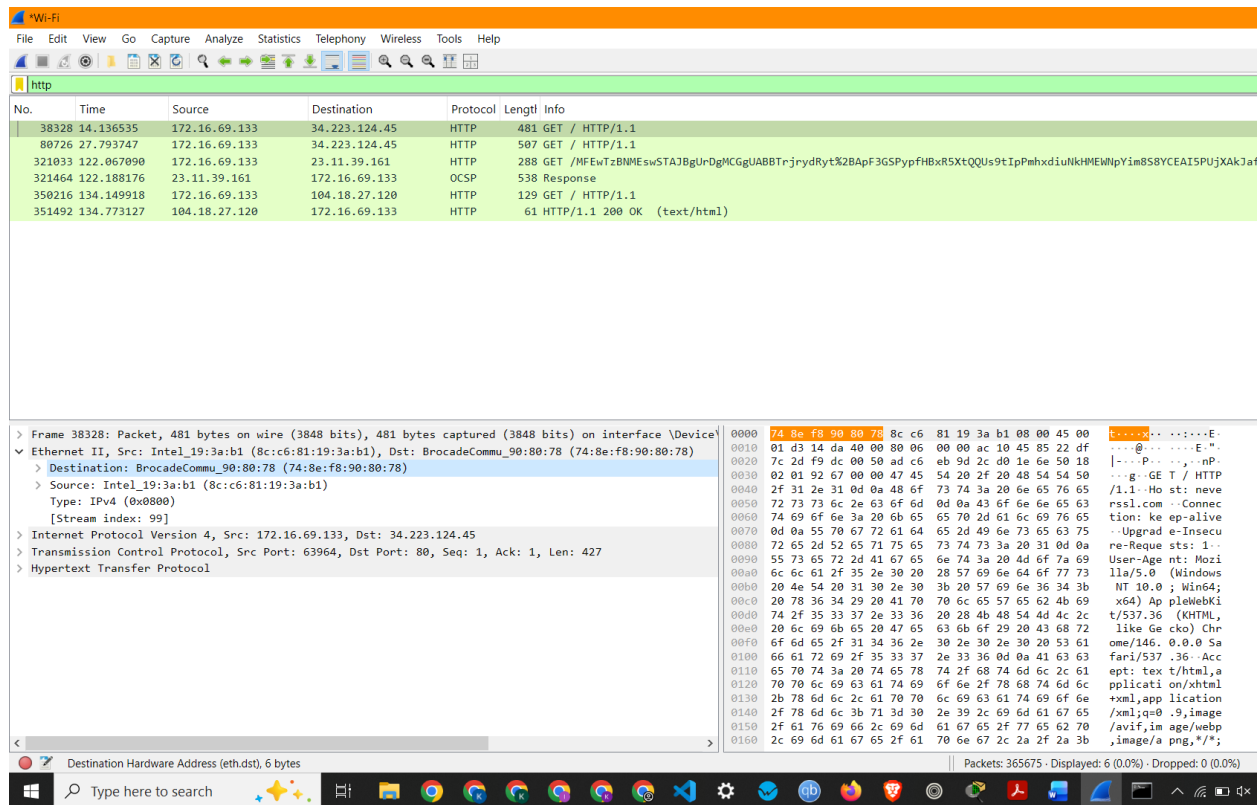
0150 2f 61 76 69 66 2c 69 6d 61 67 65 2f 77 65 62 70 /avif,image/webp

0160 2c 69 6d 61 67 65 2f 61 70 6e 67 2c 2a 2f 2a 3b ,image/a png,*/*;

0170 71 3d 30 2e 38 2c 61 70 70 6c 69 63 61 74 69 6f q=0.8,ap plicatio

Packets: 365675 - Displayed: 6 (0.0%) - Dropped: 0 (0.0%)

Screenshot showing Source and Destination IP Address of a HTTP



Screenshot showing Source and Destination MAC Address of a HTTP Packet

Observation: HTTP Packet 38328 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.16.69.133
- Destination IP Address: 34.223.124.45
- Source MAC Address: 8c:c6:81:19:3a:b1
- Destination MAC Address: 74:8e:f8:90:80:78
- Packet Length: 481 Bytes